

Investigating the Formation of Heavy Double Neutron Stars

Is the formation of GW190425 really mysterious?

Pulsar Coffee update

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Supervisors : Dr. Simon Stevenson and Prof. Jarrod Hurley

Radio Telescopes



Parkes Radio Observatory

~ 26 Galactic DNS in the field

Total mass: $2.5 - 2.9 M_{\odot}$

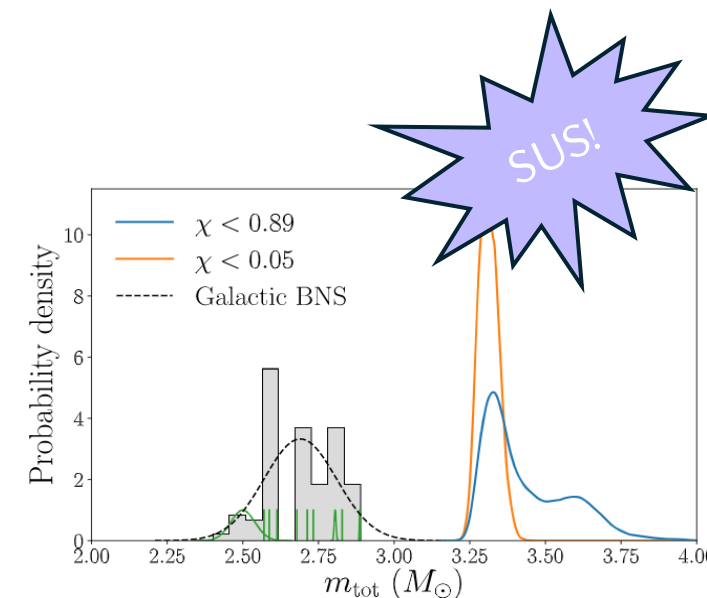
Gravitational Wave Observatories



GW170817: Total mass : $\sim 2.8 M_{\odot}$

GW190425: Total mass : $\sim 3.4 M_{\odot}$

GW231109_235456 : $\sim 2.95 M_{\odot}$ (if astrophysical!)



Abbot et al. 2020

DNS Total mass $\geq 3 M_{\odot}$: Heavy DNSs

Believed to form through the standard formation channel (T. Tauris et al. 2017)

Alternate formation channels/pathways for the formation of GW190425

Basically, similar formation pathway just different initial conditions and following mass transfer processes!

No self-consistent, unbiased formation pathway to explain heavy double neutron star formation!

My role : Test all the formation channels and find the winner!

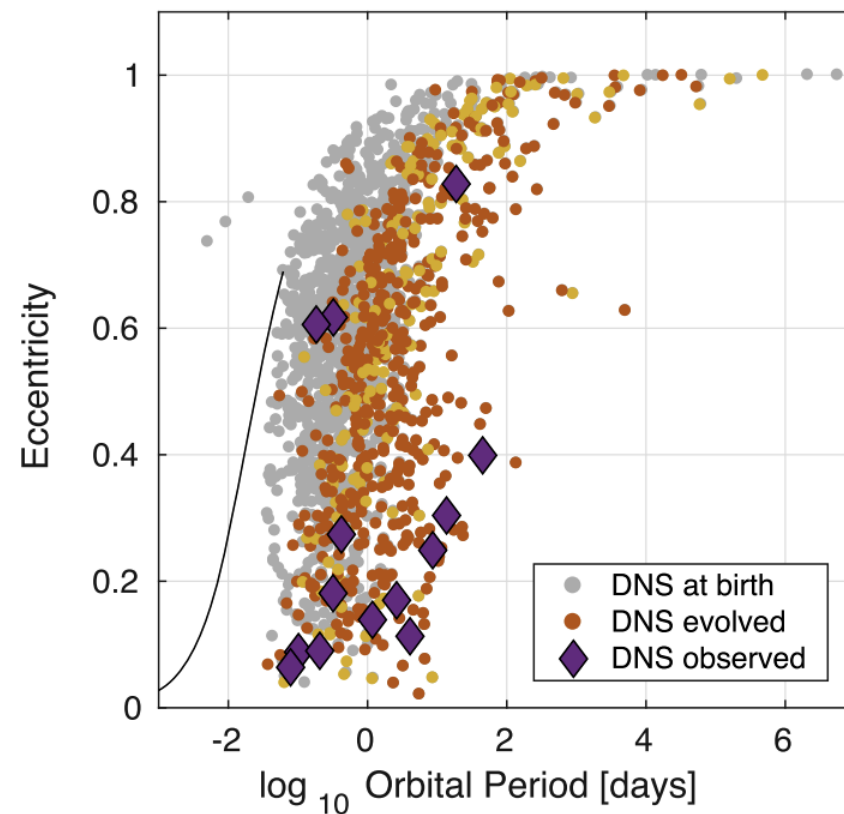
Common approach for population studies

Rapid Binary Population Synthesis

e.g., COMPAS (Stevenson et al. 2017),
COMBINE (Kruckow et al. 2018)

Caveat : Fail to reproduce the Galactic DNS properties!

e.g., Orbital period – eccentricity distribution



Vigna Gómez et al. (2018)

(Recap) ***Project I, Part A***

Simulation of Double Neutron Stars

(using detailed stellar evolution code MESA + postprocessing framework)

Initial parameter space

He ZAMS mass: $2.5 - 10 M_{\odot}$

First-born NS (NS1) mass: $1.4 M_{\odot}$

Orbital period: 1.5 hrs – 100 days

Metallicity: $Z = \text{Solar metallicity } (Z_{\odot})$

we could not form enough heavy DNSs to explain the observation of GW190425-like systems and compare all the formation channels

Project I, Part B

Extended Initial parameter space

He ZAMS : $2.5 - 10 M_{\odot}$



First-born NS (NS1) : $1.1 - 1.9 M_{\odot}$

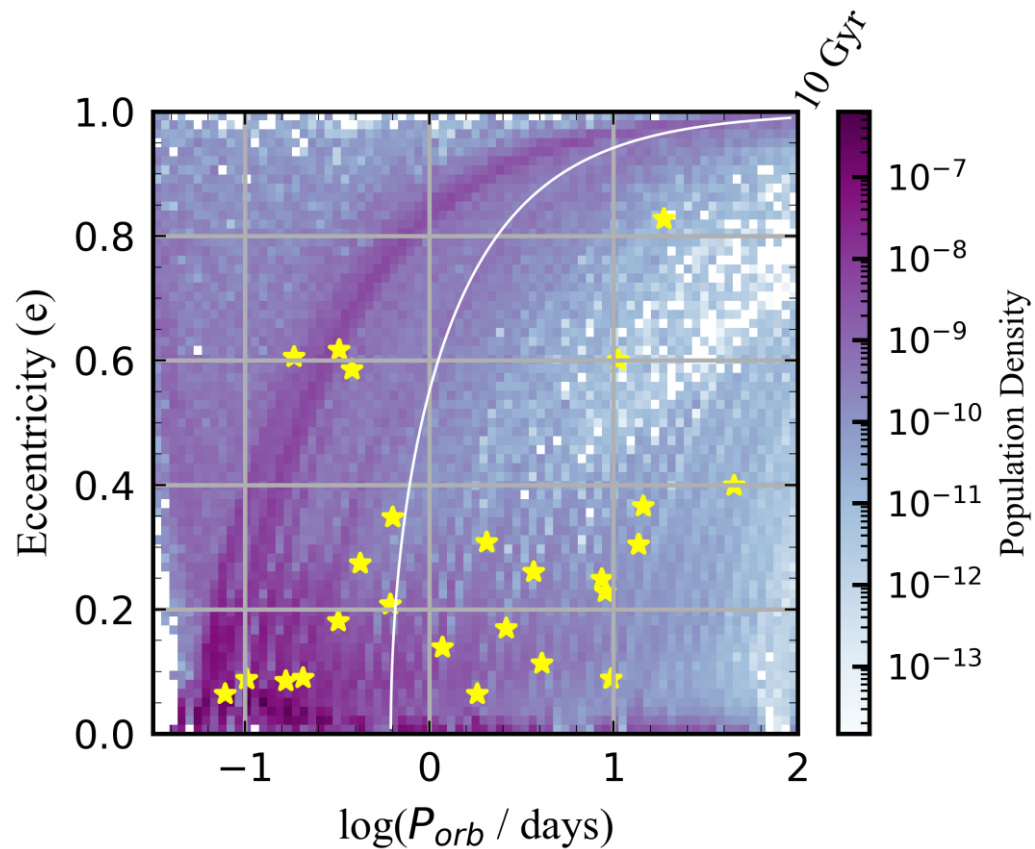
Orbital period : 1.5 hrs – 100 days



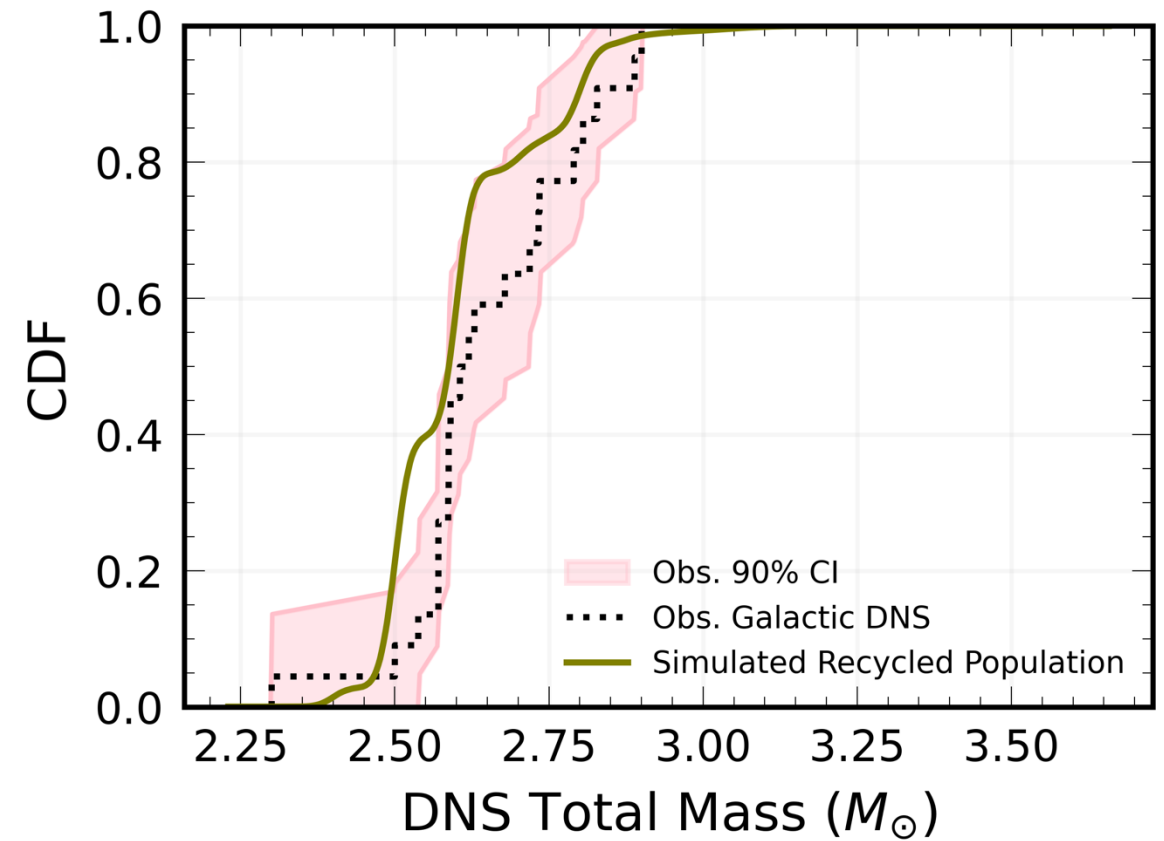
Metallicity : $Z = Z_{\odot}$, $Z = 0.1 Z_{\odot}$ and $Z = 0.01 Z_{\odot}$

Simulated DNS population compared against observed DNS population ($Z = Z_{\odot}$)

Combined Porb-e Distribution of all models

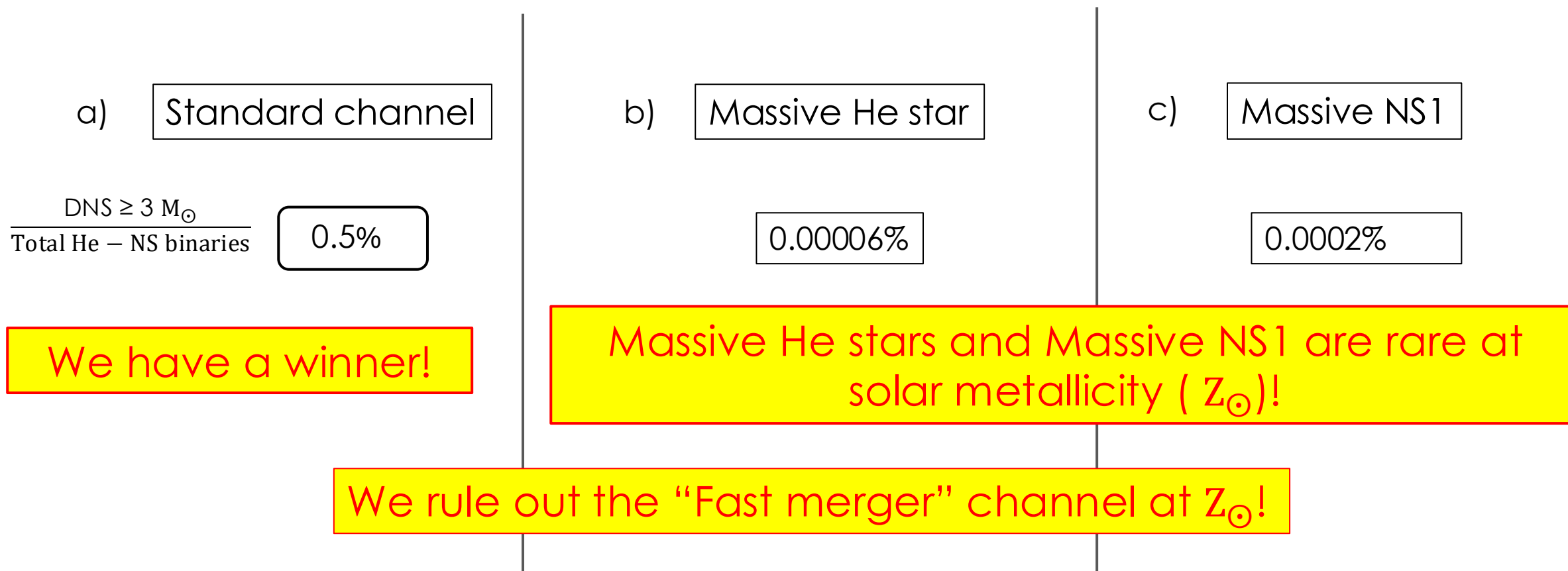


DNS Total Mass Distribution



Formation of Double Neutron Stars at Solar Metallicity

GW190425 Progenitors



Formation of Double Neutron Stars at Low Metallicity

Metallicity (Z) also plays a crucial role in shaping the properties of compact remnants.

One of the important metallicity-dependent effects in massive stars is mass loss through radiation (line)-driven (radiation transfers momentum to metal ions through spectral-line absorption, driving material away from the stellar surface hence line driven. Note: Why metals are important? H and He do not have as many spectral lines in the ultraviolet (UV) that can efficiently absorb the star's radiation and transfer momentum to the gas compared to that from metal ions).

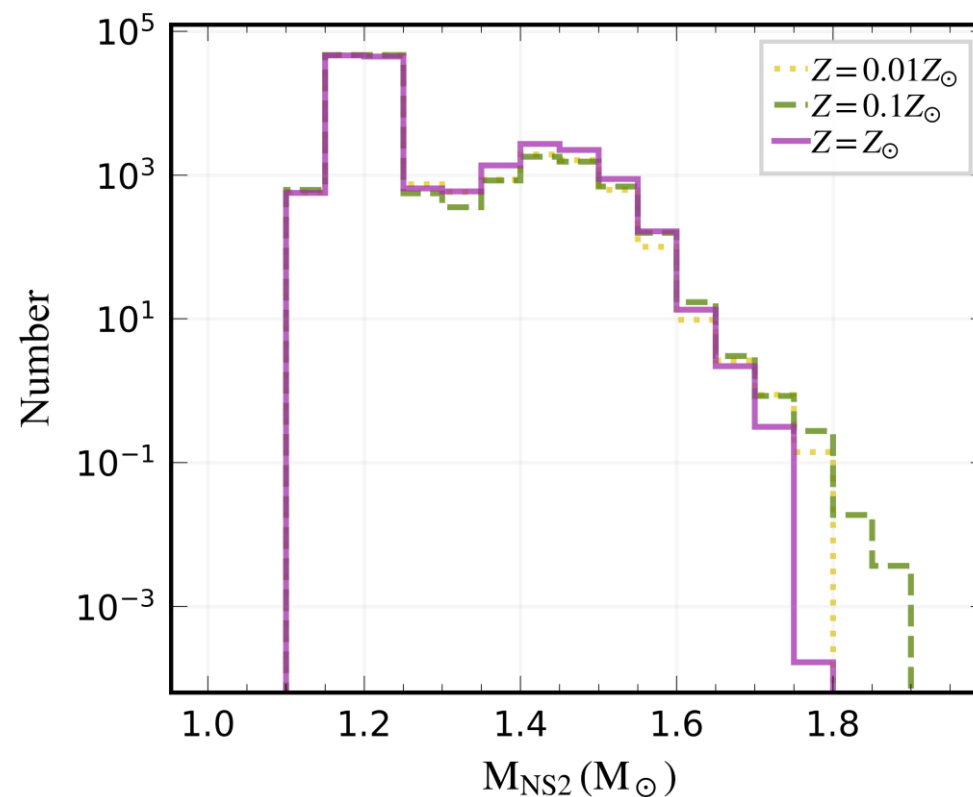
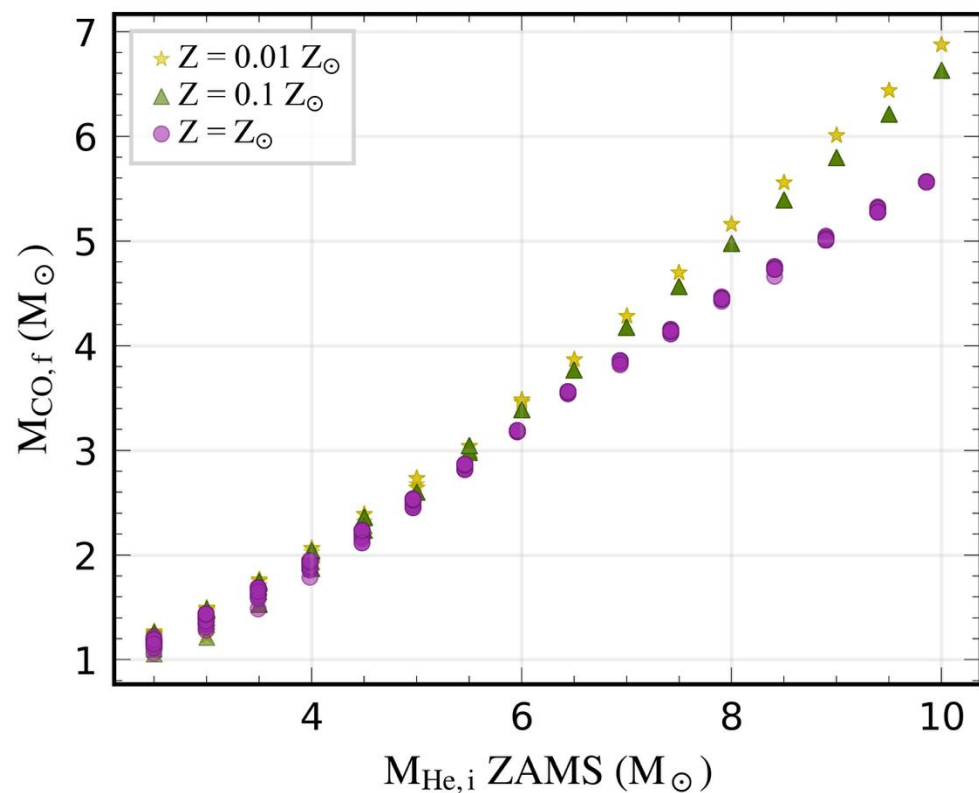
Therefore,

$Z \uparrow \Rightarrow$ more metal ions/lines $\Rightarrow$ stronger radiative acceleration $\Rightarrow \uparrow$ in wind mass loss rates

$Z \downarrow \Rightarrow \downarrow$ in wind mass loss rates

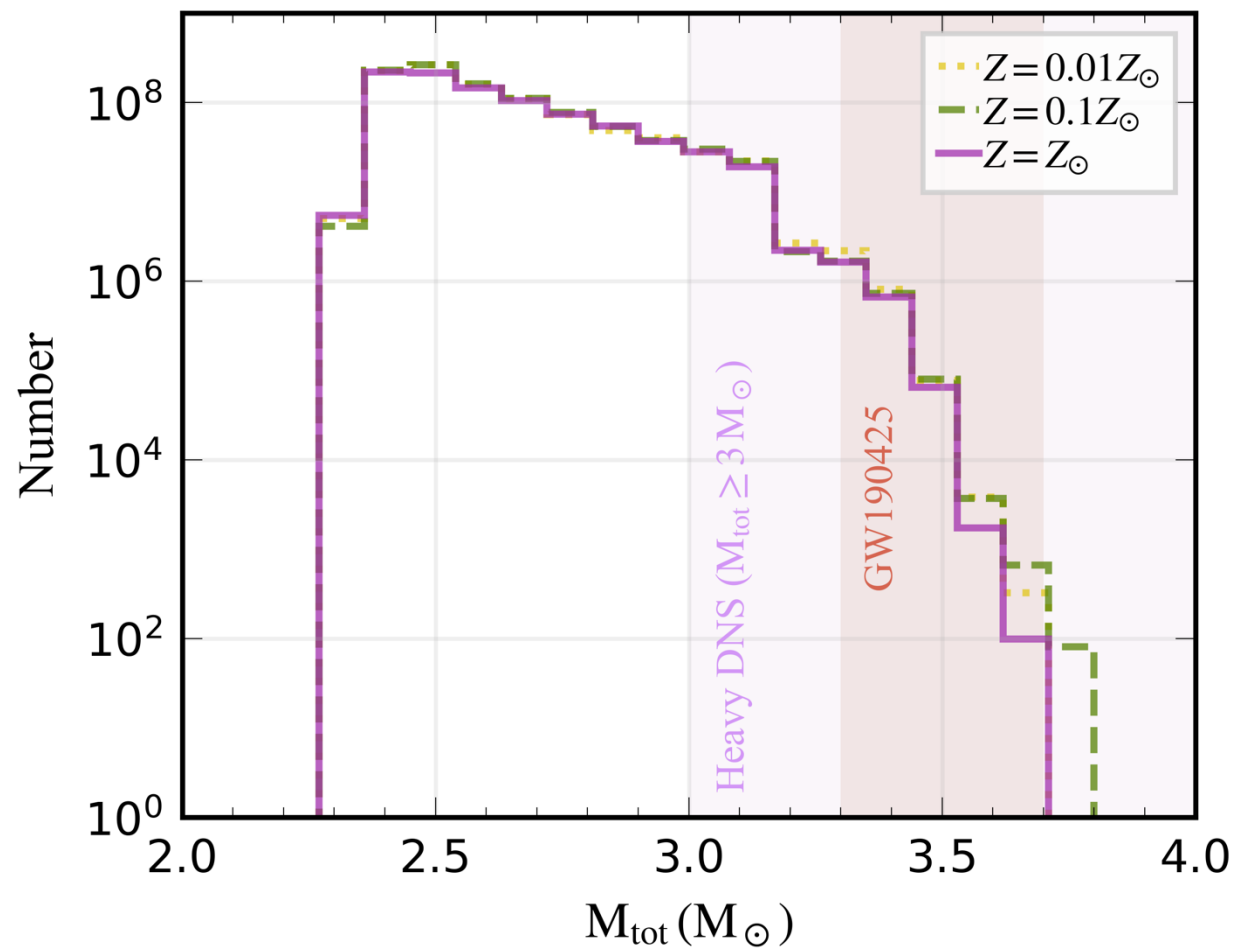
Formation of Double Neutron Stars at Low Metallicity

At lower metallicities, weaker winds lead to reduced mass loss, allowing stars to retain more mass prior to core collapse and form a more massive remnant



Formation of Double Neutron Stars at Low Metallicity

GW190425 Progenitors



0.5% more than $Z = Z_{\odot}$

1. Choice of the remnant mass prescription that maps pre-SN co metal core mass to remnant mass

SN mass loss: changes how much mass is ejected during the explosion

Kick physics, Binary survival: affects whether the system remains bound after the SN

NS2 mass: changes the mass distribution of the second-born NS

Post-SN orbital parameters: changes orbital period and eccentricity

DNS merger time: affects whether the system merges within a Hubble time

Total DNS mass: affects the fraction of heavy DNS systems ($M_{\text{tot}} \gtrsim 3 M_{\odot}$)

DNS formation efficiency: changes the fraction of progenitors that successfully form DNSs

DNS/BHNS boundary: for massive cores, can affect whether the remnant is an NS or BH

2. Maybe metallicity has a stronger impact during the initial evolutionary phases?

- During the H-rich phase => metallicity modifies stellar radial expansion, age of RLOF mass transfer => affecting how long the envelope stripping lasts
- **At solar metallicity** => RLOF occurs shortly after the MS phase => star's envelope expands rapidly, mass transfer begins relatively early => allows the stripping of the envelope to last longer.
- **At low metallicity** => stars do not expand enough to overflow the RL immediately after the MS phase => RLOF later during core-He burning => shortens the duration of the stripping of the star.
- The uncertainties associated with the common envelope evolution add to the puzzle of the formation of DNSs.
- Significantly alter properties of He star entering the He star + NS grid compared to He stars at solar metallicity.
- This will significantly alter the mass and orbital separation parameter space of He star + NS binaries that become heavy DNSs.

3. Better understanding of stellar winds from the progenitor He-rich stripped stars

- Metallicity variation can affect the hydrogen mass fraction retained onto the helium core => can further affect the mass that the He star loses via wind loss, radius evolution, mass transfer cases boundaries, final CO core mass, remnant mass.
- The metallicity effect is important, but it might still be weaker in structurally modifying the He star than in an H-rich star.

Takeaways

Maybe we don't need a new formation channel to explain the formation of GW190425!

Weak dependence of metallicity on the formation of heavy systems (only 0.5% more systems)

Need more observations and detailed studies of the late phase binary evolution

Formation of heavy double neutron stars — I. Eddington-limited accretion for a $1.4 M_{\odot}$ neutron star at solar metallicity

Show affiliations

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Astrophysics > Solar and Stellar Astrophysics

[Submitted on 18 Aug 2026]

Formation of heavy double neutron stars II: the role of heavy first-born neutron stars and low metallicity

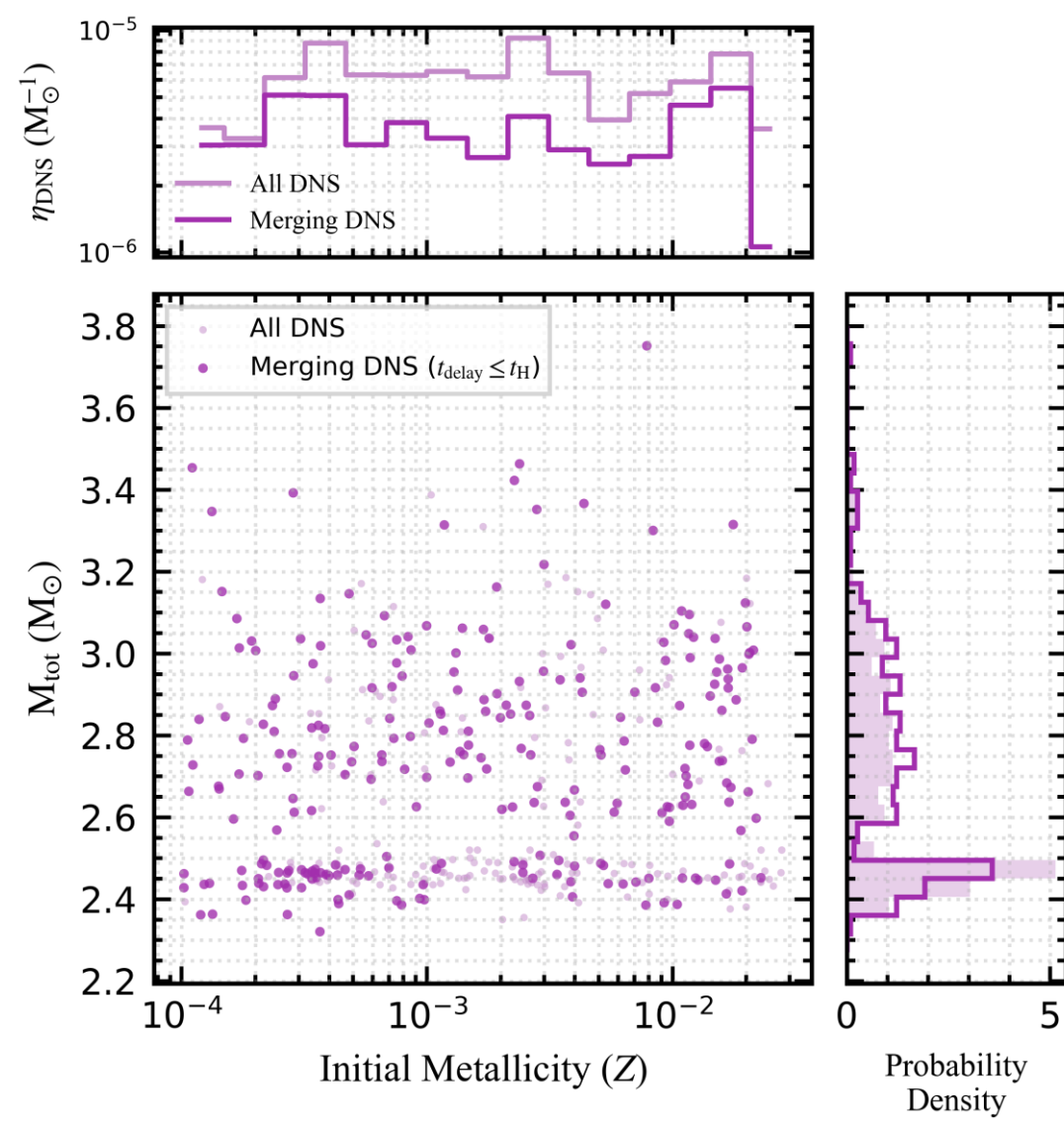
Ashwathi Nair, Simon Stevenson



Visit to Geneva Observatory, Switzerland

POSYDON School- **POSYDON** (**PO**population **SY**nthesis with **D**etailed binary-evolution simulations) : single and binary-star population synthesis code incorporating full stellar structure and evolution modeling with the use of MESA.

Visit to Institute of Science and Technology Austria (Ylva Gotberg's group)



Discovering the Origins of Gravitational Wave Sources

Ashwathi Nair (Final Year PhD – 100% looking for a Postdoc:))

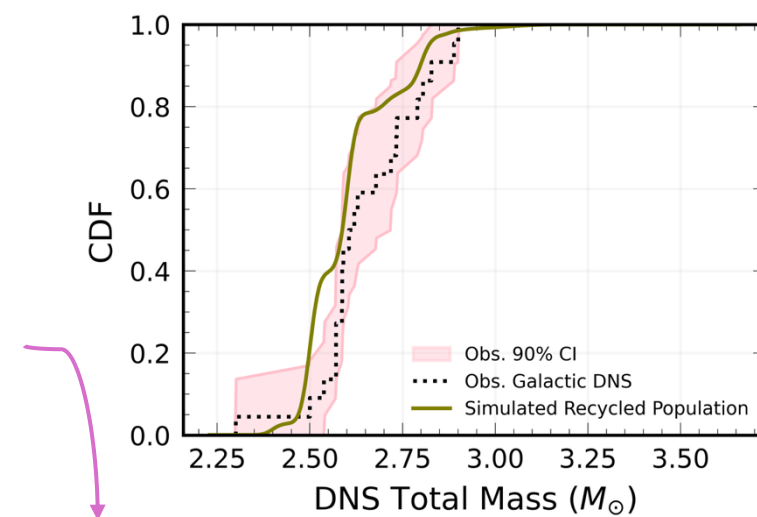
Supervisors: Dr. Simon Stevenson and Prof. Jarrod Hurley



Major interests : Stellar astrophysics: Binary star evolution, progenitor of gravitational wave sources: formation and evolution and detailed studies of physical processes that decides the final fate of these binaries!

Few questions I spent addressing during my last three years:

1. Can we explain the population of observed radio Galactic DNSs and Gravitational Wave DNS mergers with our detailed models?
2. Sure, GW190425 is a sus (with a total mass of $3.4M_{\text{sun}}$), but do we need different formation scenarios to explain its formation and such heavy system's absence in radio population? Is the formation of **heavy double neutron star (DNS) GW190425** really mysterious?

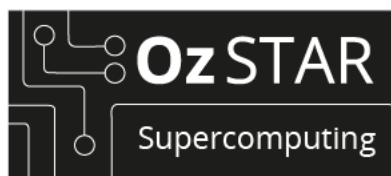


Tools used:

MESA

Paxton et al. 2011

Big Thanks
to:



Read the complete story here:



Part I



Part II

We can closely explain it, and we find that

1. Formation of heavy GW190425 like DNSs are rare at solar metallicity !
2. And no, we do not need fancy formation channels to explain its formation. It can be explained with the **og standard isolated formation channel!!**